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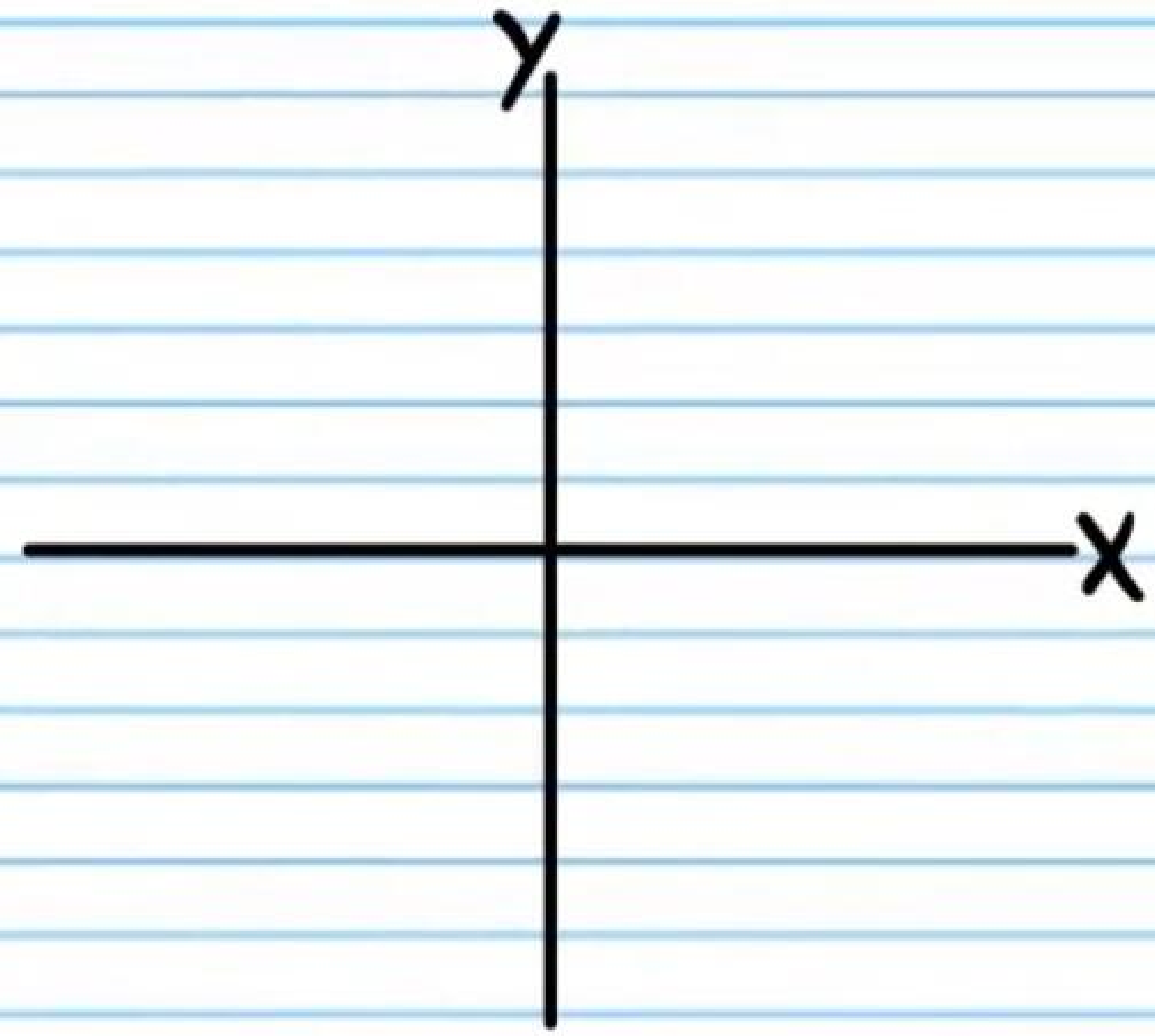
Using the Pythagorean Theorem to Find the Values of Trigonometric Functions (Homework)

① If $\sin\theta = -\frac{3}{5}$, and θ terminates in quadrant III, find the values of the five remaining basic trigonometric functions of θ.



i) $\csc\theta$

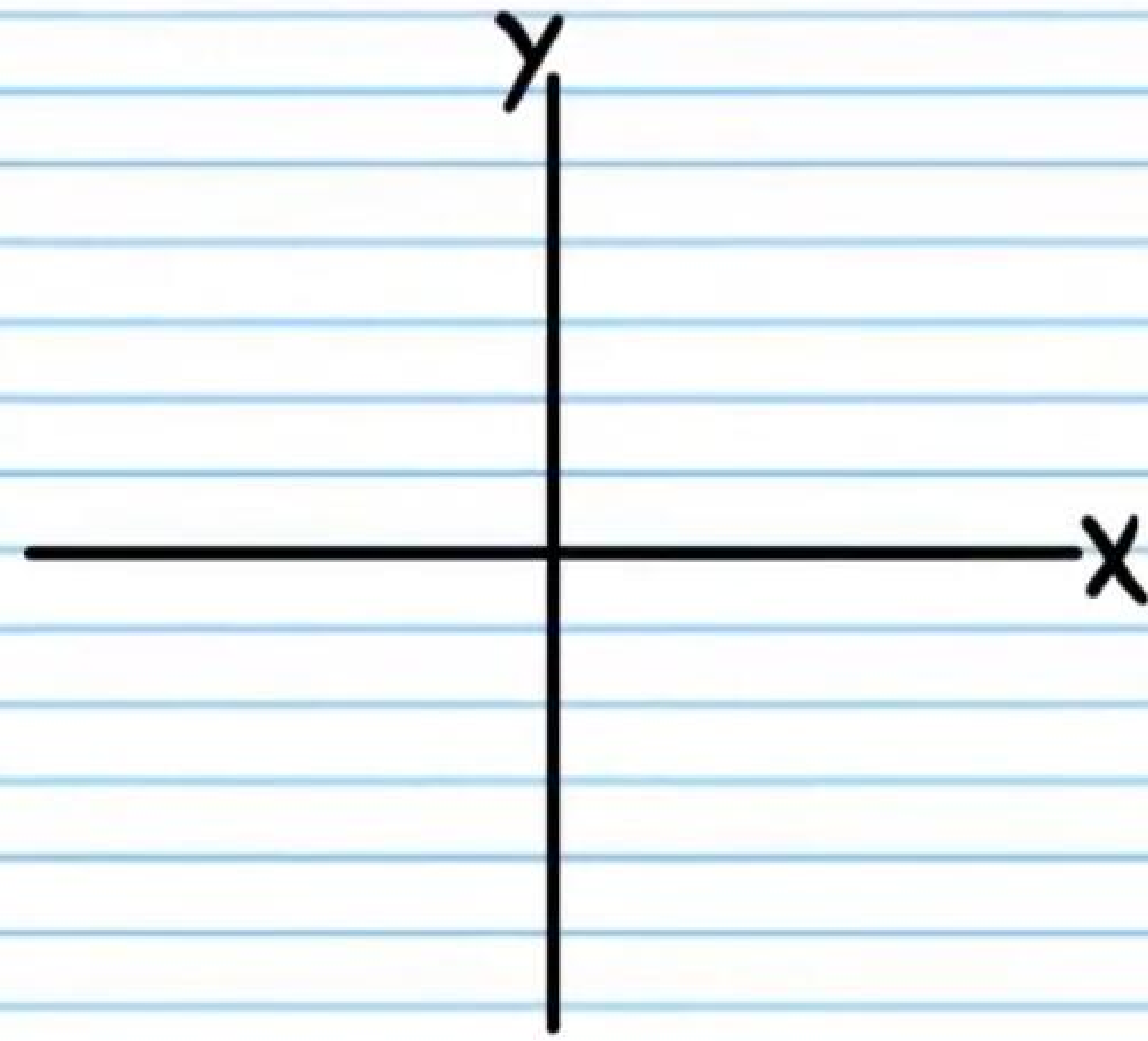
ii) $\sec\theta$

iii) $\cot\theta$

iv) $\cos\theta$

v) $\tan\theta$

② If $\tan W = -\frac{1}{3}$, and W terminates in quadrant II, find the values of the five remaining basic trigonometric functions of W.



i) $\csc W$

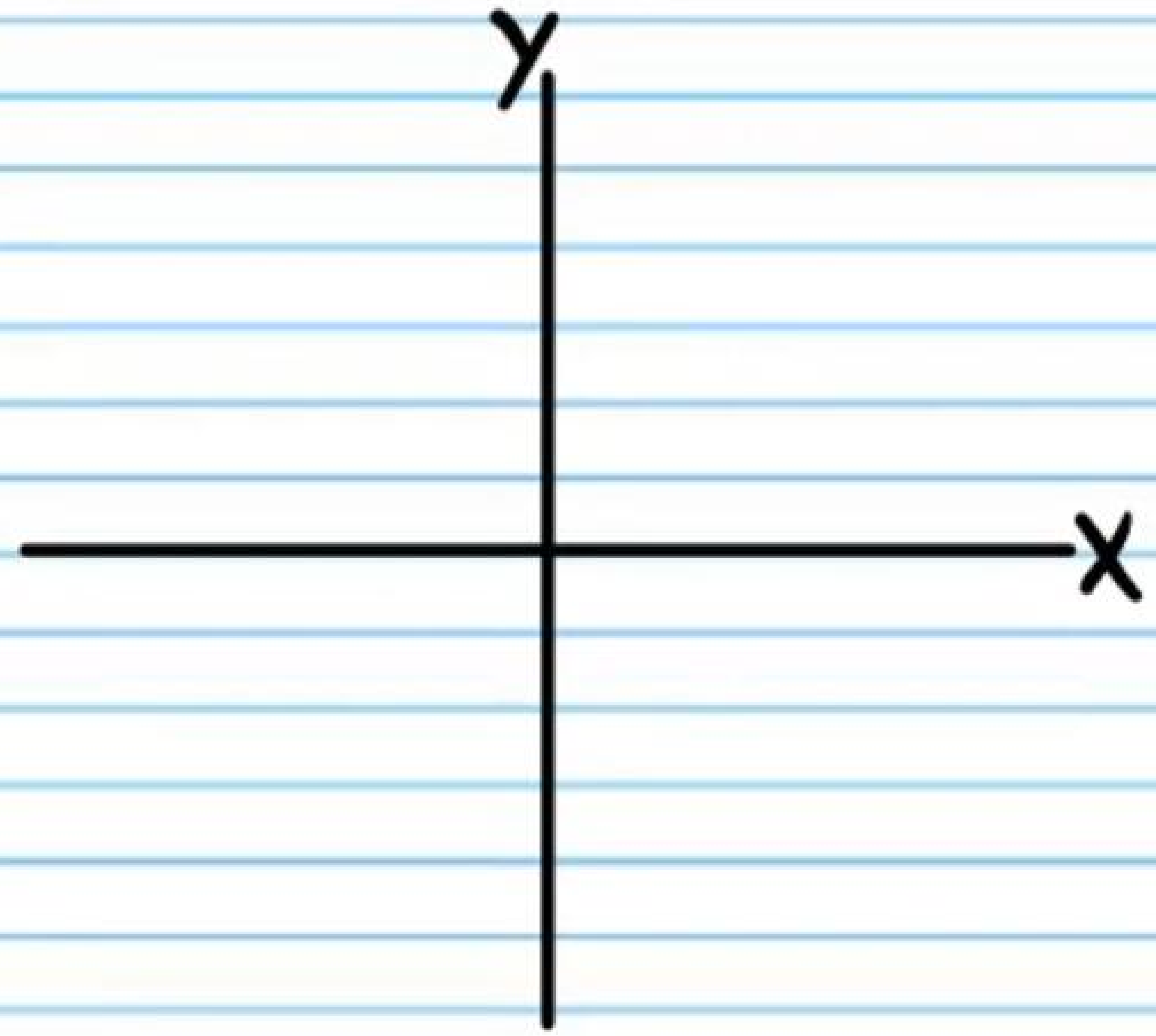
ii) $\sec W$

iii) $\cot W$

iv) $\cos W$

v) $\sin W$

③ If $\sec\alpha = 4$, and α terminates in quadrant I, find the values of the five remaining basic trigonometric functions of α.



i) $\csc\alpha$

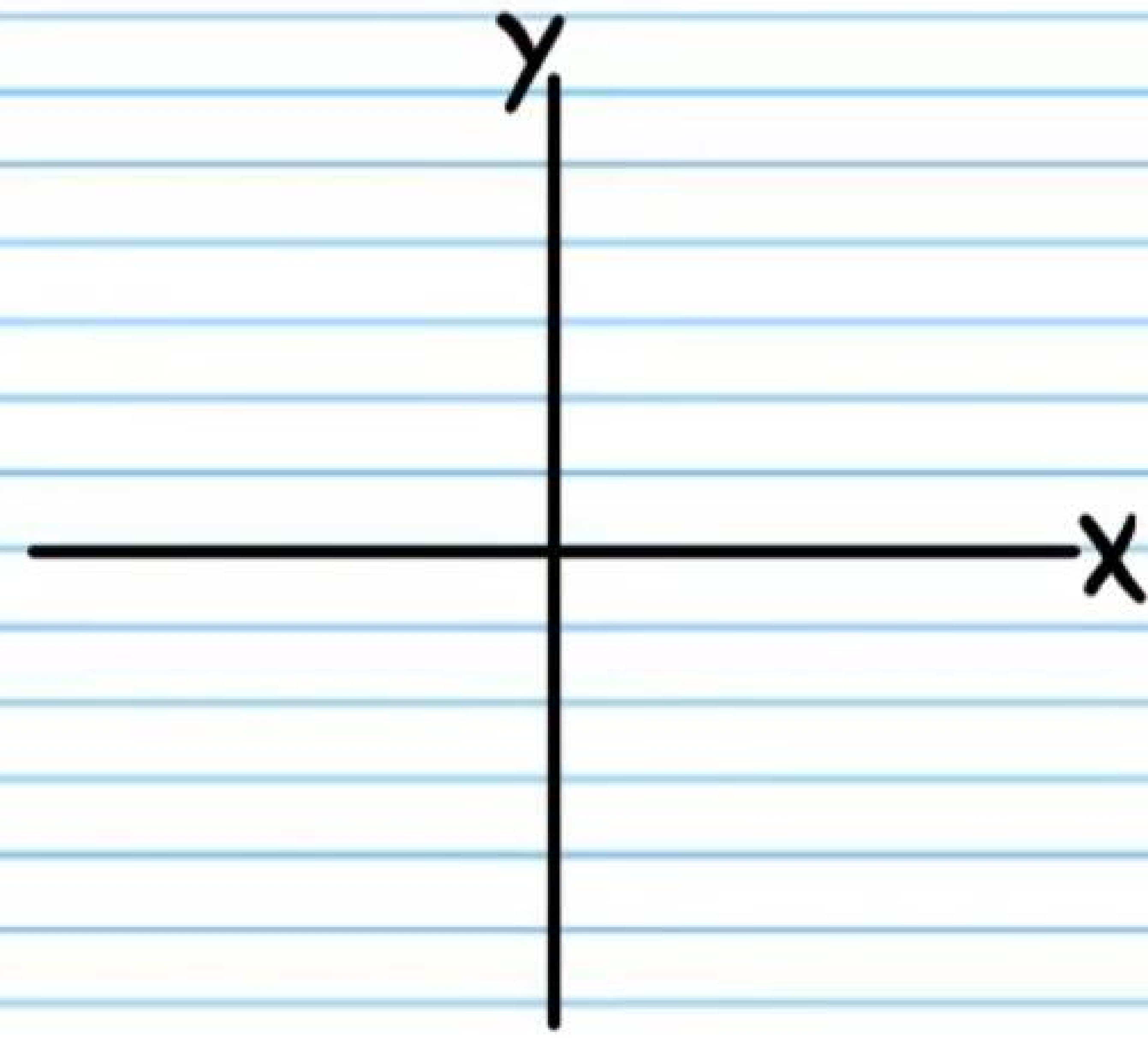
ii) $\sin\alpha$

iii) $\cot\alpha$

iv) $\cos\alpha$

v) $\tan\alpha$

④ If $\cos\beta = \frac{8}{9}$, and β terminates in quadrant IV, find the values of the five remaining basic trigonometric functions of β.



i) $\csc\beta$

ii) $\sec\beta$

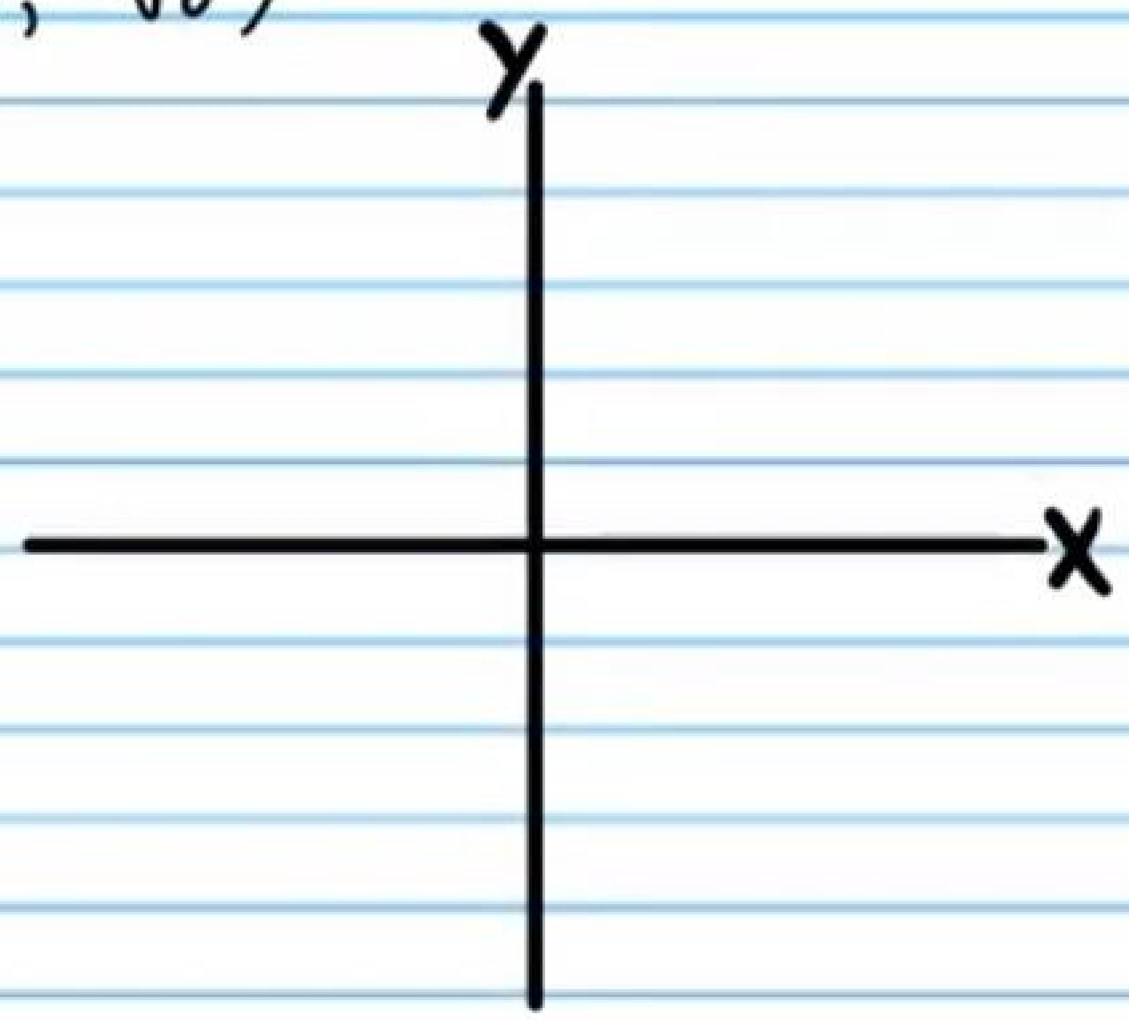
iii) $\cot\beta$

iv) $\sin\beta$

v) $\tan\beta$

The coordinates of points are given below. If each of these points lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

5 $(2, -\sqrt{6})$



i) $\sin\theta$

ii) $\cos\theta$

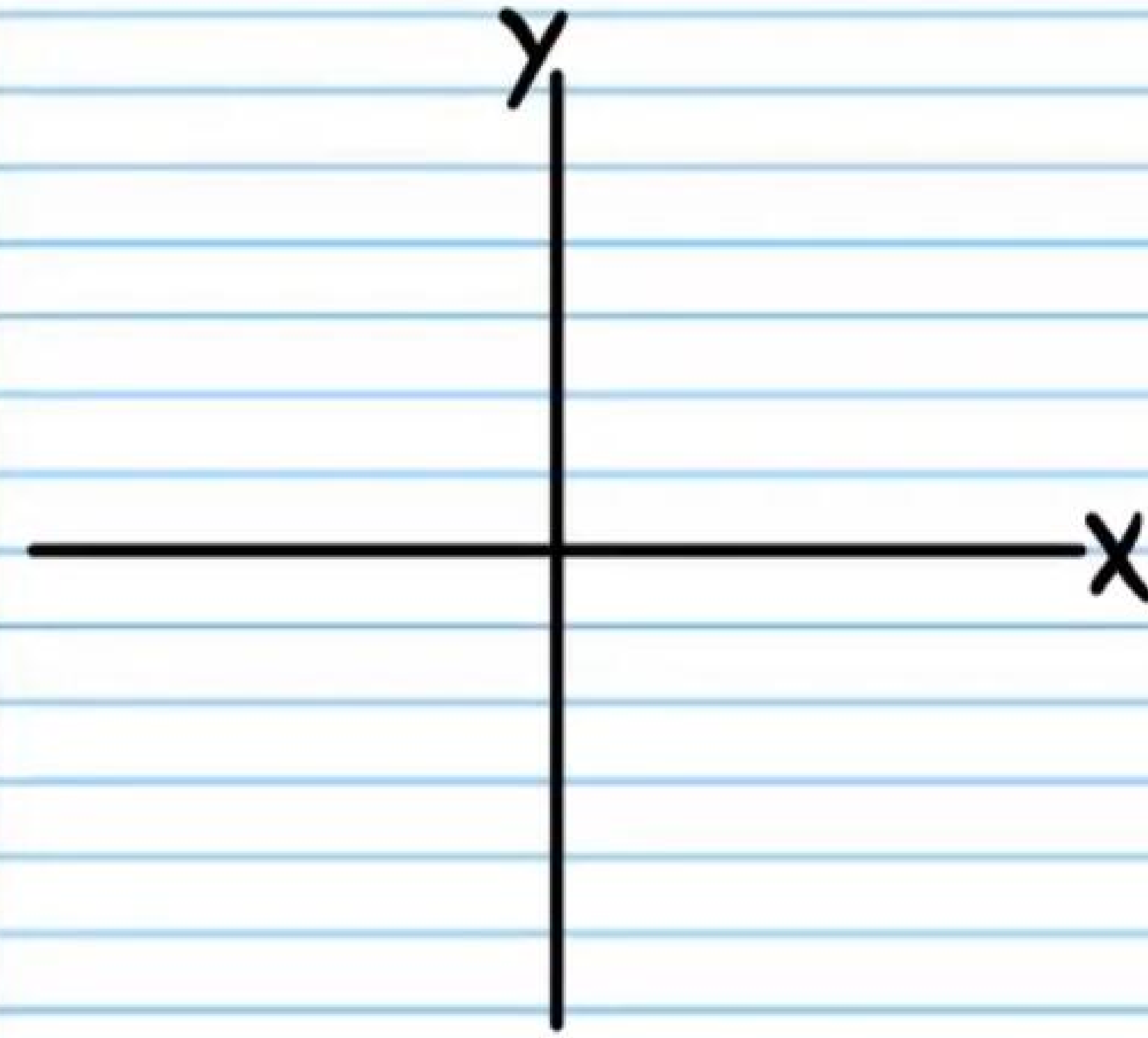
iii) $\tan\theta$

iv) $\csc\theta$

v) $\sec\theta$

vi) $\cot\theta$

6 $(-7, 7)$



i) $\sin\theta$

ii) $\cos\theta$

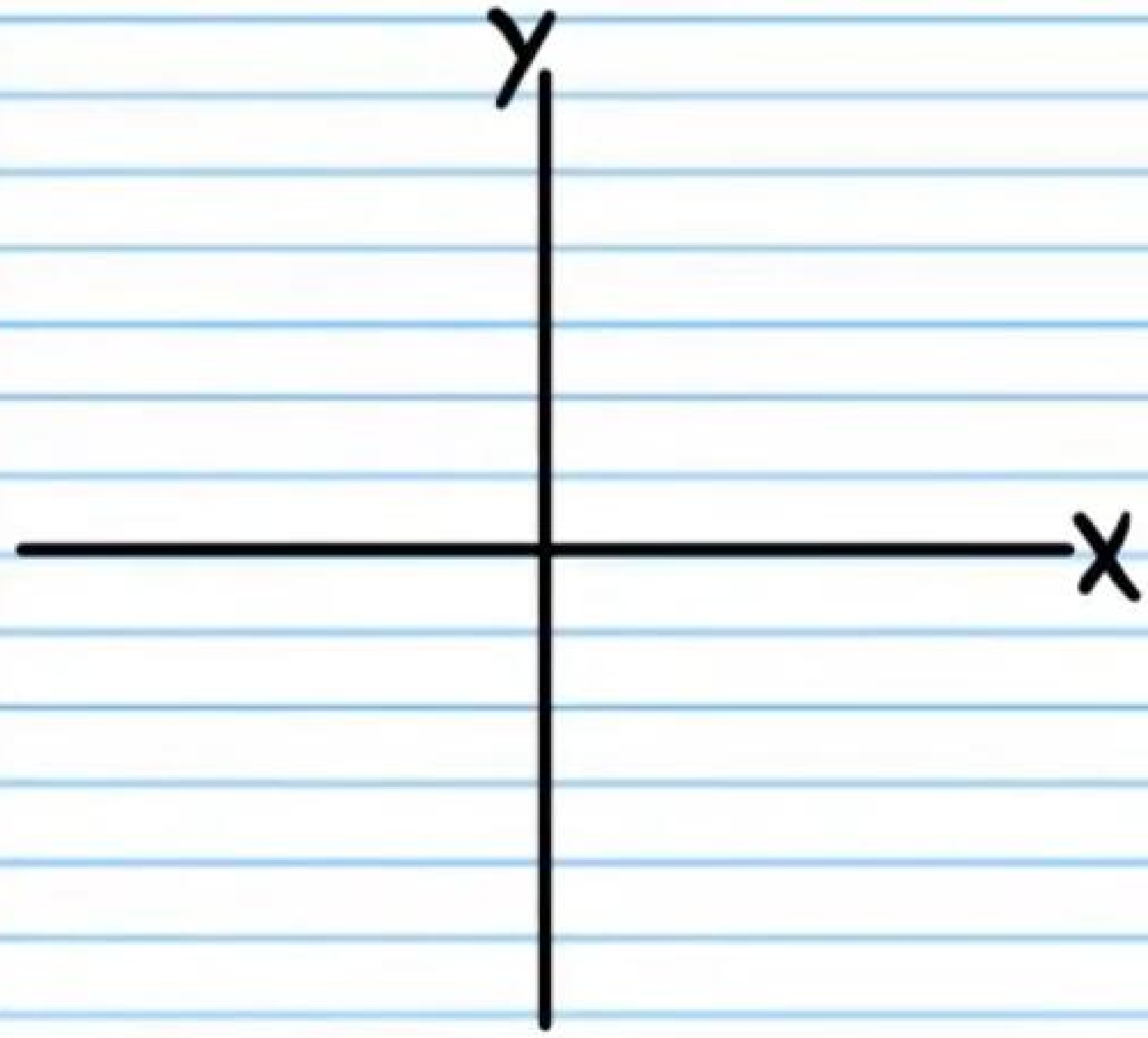
iii) $\tan\theta$

iv) $\csc\theta$

v) $\sec\theta$

vi) $\cot\theta$

⑦ $(-4, -2)$



i) $\sin\theta$

ii) $\cos\theta$

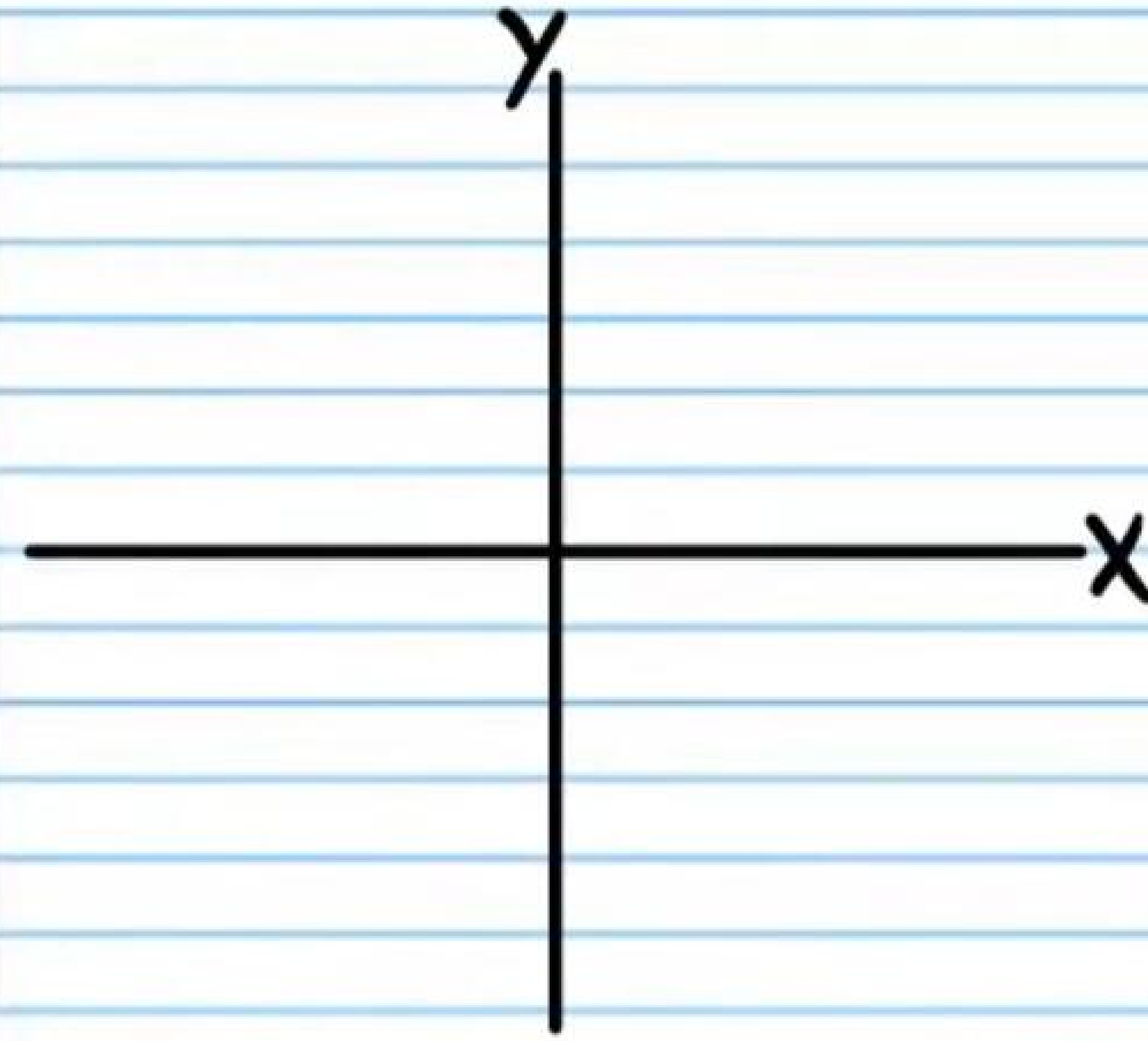
iii) $\tan\theta$

iv) $\csc\theta$

v) $\sec\theta$

vi) $\cot\theta$

⑧ $(6, 6\sqrt{3})$



i) $\sin\theta$

ii) $\cos\theta$

iii) $\tan\theta$

iv) $\csc\theta$

v) $\sec\theta$

vi) $\cot\theta$

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Homework Answers

① i) $\csc\theta = \boxed{-\frac{5}{3}}$

ii) $\sec\theta = \boxed{-\frac{5}{4}}$

iii) $\cot\theta = \boxed{\frac{4}{3}}$

iv) $\cos\theta = \boxed{-\frac{4}{5}}$

v) $\tan\theta = \boxed{\frac{3}{4}}$

② i) $\csc w = \boxed{\sqrt{10}}$

ii) $\sec w = \boxed{-\frac{\sqrt{10}}{3}}$

iii) $\cot w = \boxed{-3}$

iv) $\cos w = -\frac{3}{\sqrt{10}} = \boxed{-\frac{3\sqrt{10}}{10}}$

v) $\sin w = \frac{1}{\sqrt{10}} = \boxed{\frac{\sqrt{10}}{10}}$

③ i) $\csc\alpha = \frac{4}{\sqrt{15}} = \boxed{\frac{4\sqrt{15}}{15}}$

ii) $\sin\alpha = \boxed{\frac{\sqrt{15}}{4}}$

iii) $\cot\alpha = \frac{1}{\sqrt{15}} = \boxed{\frac{\sqrt{15}}{15}}$

iv) $\cos\alpha = \boxed{\frac{1}{4}}$

v) $\tan\alpha = \boxed{\sqrt{15}}$

④ i) $\csc\beta = -\frac{9}{\sqrt{17}} = \boxed{-\frac{9\sqrt{17}}{17}}$

ii) $\sec\beta = \boxed{\frac{9}{8}}$

iii) $\cot\beta = -\frac{8}{\sqrt{17}} = \boxed{-\frac{8\sqrt{17}}{17}}$

iv) $\sin\beta = \boxed{-\frac{\sqrt{17}}{9}}$

v) $\tan\beta = \boxed{-\frac{\sqrt{17}}{8}}$

⑤ i) $\sin\theta = -\frac{\sqrt{6}}{\sqrt{10}} = -\frac{\sqrt{60}}{10} = -\frac{2\sqrt{15}}{10} = \boxed{-\frac{\sqrt{15}}{5}}$

ii) $\cos\theta = \frac{2}{\sqrt{10}} = \frac{2\sqrt{10}}{10} = \boxed{\frac{\sqrt{10}}{5}}$

iii) $\tan\theta = \boxed{-\frac{\sqrt{6}}{2}}$

iv) $\csc\theta = -\frac{\sqrt{10}}{\sqrt{6}} = -\frac{\sqrt{60}}{6} = -\frac{2\sqrt{15}}{6} = \boxed{-\frac{\sqrt{15}}{3}}$

v) $\sec\theta = \boxed{\frac{\sqrt{10}}{2}}$

vi) $\cot\theta = -\frac{2}{\sqrt{6}} = -\frac{2\sqrt{6}}{6} = \boxed{-\frac{\sqrt{6}}{3}}$

⑥ i) $\sin\theta = \boxed{\frac{\sqrt{2}}{2}}$

ii) $\cos\theta = \boxed{-\frac{\sqrt{2}}{2}}$

iii) $\tan\theta = \boxed{-1}$

iv) $\csc\theta = \boxed{\sqrt{2}}$

v) $\sec\theta = \boxed{-\sqrt{2}}$

vi) $\cot\theta = \boxed{-1}$

⑦ i) $\sin\theta = \boxed{-\frac{\sqrt{5}}{5}}$

ii) $\cos\theta = \boxed{-\frac{2\sqrt{5}}{5}}$

iii) $\tan\theta = \boxed{\frac{1}{2}}$

iv) $\csc\theta = \boxed{-\sqrt{5}}$

v) $\sec\theta = \boxed{-\frac{\sqrt{5}}{2}}$

vi) $\cot\theta = \boxed{2}$

⑧ i) $\sin\theta = \frac{6\sqrt{3}}{12} = \boxed{\frac{\sqrt{3}}{2}}$

ii) $\cos\theta = \frac{6}{12} = \boxed{\frac{1}{2}}$

iii) $\tan\theta = \frac{6\sqrt{3}}{6} = \boxed{\sqrt{3}}$

iv) $\csc\theta = \frac{12}{6\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$

v) $\sec\theta = \frac{12}{6} = \boxed{2}$

vi) $\cot\theta = \frac{6}{6\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$